Bartłomiej Baranowski

PhD Candidate, Visual Computing Lab, Technical University of Munich

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I work on spatial intelligence for perceiving, reasoning about, and acting in the 3D world, spanning 3D reconstruction, scene understanding, and vision-language models.

Education

Technical University of Munich — *Nov 2025 – present*
PhD Candidate, Visual Computing Lab
- Advised by Prof. Dr. Matthias Nießner.

University of Tübingen — *2022 – 2025*
MSc in Machine Learning
- **Thesis** and **research project** in Prof. A. Geiger's group on 3D Gaussian Splatting, improving densification and initialization with elements of neural radiance fields and reducing storage cost. Research assistant in the group, Sep 2025 – Nov 2025.
- Research assistant in Prof. Baumgartner's group (medical imaging), Apr 2023 – Dec 2023.

Lodz University of Technology — *2018 – 2022*
BEng in Computer Science

Publications

OneCanvas: 3D Scene Understanding via Panoramic Reprojection
B. Baranowski, D. Z. Chen, M. Nießner. arXiv preprint, 2026.

ConeGS: Error-Guided Densification Using Pixel Cones for Improved Reconstruction with Fewer Primitives
B. Baranowski, S. Esposito, P. Gschoßmann, A. Chen, A. Geiger. International Conference on 3D Vision (3DV), 2026. ORAL

Experience

Data Scientist — *Jan 2022 – Oct 2022*
MIM Solutions (Folliscan)
- Ultrasound-based assessment of ovarian reserve, detecting and counting antral follicles from ultrasound video with 2D and 3D deep learning models (PyTorch) served through an API.

Software Developer — *Aug 2021 – Jan 2022*
Digica
- Object detection and semantic segmentation of street-utility objects from UAV footage (Python, PyTorch).

Associate Software Developer — *Dec 2019 – Jan 2021*
Fujitsu Łódź
- Automotive digital-twin application with stream processing and object detection and similarity for vehicle video, deployed on AWS (Python, TensorFlow, Java).

Teaching

Teaching Assistant — *2026 – present*
Advanced Deep Learning for Computer Vision (ADL4CV), TUM

Awards & Honors

- Oral Presentation, 3DV 2026 (ConeGS).
- Rector's Scholarship (twice), Lodz University of Technology.

Skills

Programming. Python (PyTorch, JAX), C++, CUDA, Java.

Languages

Polish (native), English (C1), German (B2).

Interests

Playing accordion, running, climbing.